

Model Verification 1: Graphical Illustration

Plot estimated and actual costs for various order sizes as a scatter graph (cost as y-axis and size as x-axis). Compute average cost for different order sizes to eliminate market noise, making sure to incorporate enough observations in each size category to eliminate the effect of market noise. Graphical illustration is the most helpful performance analysis for clients, although it is the least helpful from a statistical perspective.

Model Verification 2: Regression Analysis

Run a regression between actual and estimated costs using all data. If the forecasting model is accurate, then the regression results should show an intercept statistically equal to zero and a slope statistically equal to one. The R^2 may be lower due to noise but the t-stat and f-value should be very high implying a suitable model. This analysis will show visually whether or not the model is working well (e.g., all order sizes). Regression is the second most useful tool to help clients evaluate our model, and the second most effective statistical technique.

Model Verification 3: Z-Score Analysis

This technique allows us to jointly evaluate both the accuracy of the market impact and timing risk models. The test consists of computing a statistical z-score to determine the number of standard deviations the actual cost was from the estimated cost. The z-score is calculated as follows:

$$Z = \frac{\text{Actual} - \text{Cost Estimated Market Impact}}{\text{Timing Risk}}$$

If the model is accurate we should find the average z-score to be close to zero and the standard deviation (or variance) to be close to 1. That is, an accurate model will have:

$$Z \sim (0, 1)$$

It is important that we compute and evaluate the z-score for various order sizes and categories such as buys and sells, market cap, volatility, and so forth to ensure the model is robust or if deficiencies exist.

The distribution of the z-score and the chi-square goodness of fit of the data test will help evaluate the model statistically. This procedure has proven the most useful tool evaluating models from both a statistical basis and real-time TCA analysis (e.g., in the algorithms or from a reporting perspective).